

ENZYMES AND ITS APPLICATIONS

Enzymes

Enzymes are macromolecular biological catalysts. They are responsible for thousands of metabolic processes that sustain life. Enzymes are highly selective catalysts, greatly accelerating both the rate and specificity of metabolic chemical reactions, from the digestion of food to the synthesis of DNA. Most enzymes are proteins, although some catalytic RNA molecules have been identified.

Definition: Enzymes are proteins specialized to catalyze biological reactions. They are among the most remarkable biomolecules because of their extraordinary specificity and catalytic power.

Mechanism of enzyme action:

The substrate binds to a specific site of the enzyme protein molecule which is called as active site. Whatever the shape of active site is, the correct substrate apparently binds uniquely in the required orientation and catalyzes the enzyme substrate reaction to form a product.



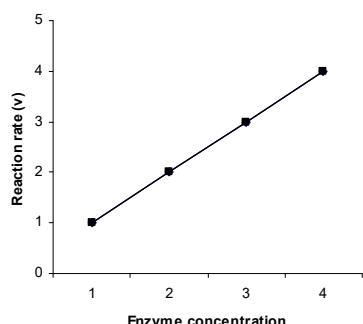
Enzymes can act in several ways, all of which lower $\Delta G^\ddagger$ (Gibbs energy):

- Lowering the activation energy by creating an environment in which the transition state is stabilized (e.g. straining the shape of a substrate—by binding the transition-state conformation of the substrate/product molecules, the enzyme distorts the bound substrate(s) into their transition state form, thereby reducing the amount of energy required to complete the transition).
- Lowering the energy of the transition state, but without distorting the substrate, by creating an environment with the opposite charge distribution to that of the transition state.
- Providing an alternative pathway. For example, temporarily reacting with the substrate to form an intermediate ES complex, which would be impossible in the absence of the enzyme.
- Reducing the reaction entropy change by bringing substrates together in the correct orientation to react.
- Increases in temperatures speed up reactions. Thus, temperature increases help the enzyme function and develop the end product even faster. However, if heated too much, the enzyme's shape deteriorates and the enzyme becomes denatured. Some enzymes like thermolabile enzymes work best at low temperatures.
- The function of a protein is dependent on its structure so that if the structure is disrupted, so is its function.

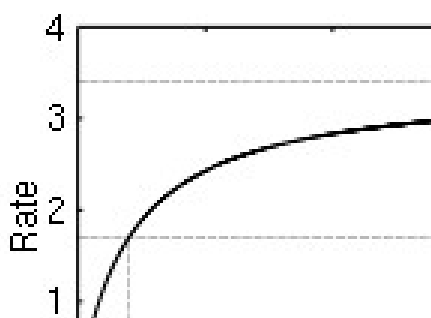
Michaelis-Menten Equation / kinetics

Michaelis Menten kinetics is one of the simplest and best known models of enzyme kinetics. It is named after German biochemist L. Michaelis and Canadian physician M.

Menten. It describes the rate of enzymatic reaction by relating reaction rate (v) to substrate concentration (S).



The rate of enzyme catalyzed reaction increases with increase in enzyme concentration but in this condition substrate should be in excess.



With fixed enzyme concentration, an increase in substrate will result in a rapid increase in rate of reaction that is called phase-I or First order kinetics. Increase in substrate will result in slow increase in rate which reaches to a constant called V_{max} at which further increase in substrate will not going to affect the rate of reaction.

At low substrate concentration the active sites on the enzyme molecules are not saturated by substrate and thus the reaction rate varies with substrate concentration (Phase-I). As the number of substrate molecules increases, the active sites are covered to a greater degree and at saturation no free active sites are available but the enzyme is working at maximum capacity (Phase-II). At this stage the rate of enzyme catalyzed reaction is independent of substrate concentration. Michaelis-Menten equation is the quantitative relationship between the rate of an enzyme catalyzed reaction and the substrate concentration. Its formula is given by

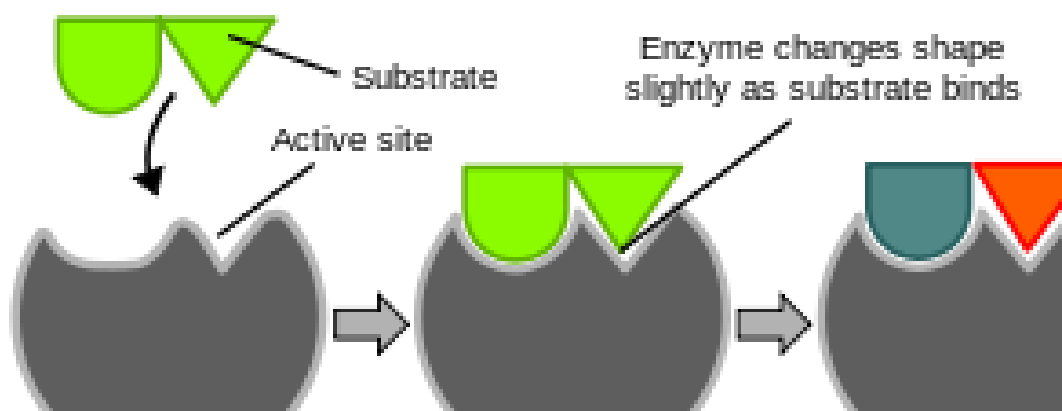
$$v = \frac{d[P]}{dt} = \frac{V}{1 + \frac{K_m}{[S]}}$$

Here, V is observed velocity at a given substrate. V_{max} represents the maximum rate achieved by the system, at maximum (saturating) substrate concentrations. The Michaelis constant K_m is the substrate concentration at which the reaction rate is half of V_{max} .

Lock and key model

Enzymes are very specific, and it was suggested by Emil Fischer in 1894 that this was because both the enzyme and the substrate possess specific complementary geometric

shapes that fit exactly into one another. This is often referred to as "the lock and key" model. However, while this model explains enzyme specificity, it fails to explain the stabilization of the transition state that enzymes achieve.



Induced fit Model

In 1958, Daniel Koshland suggested a modification to the lock and key model: since enzymes are rather flexible structures, the active site is continuously reshaped by interactions with the substrate as the substrate interacts with the enzyme. As a result, the substrate does not simply bind to a rigid active site; the amino acid side-chains that make up the active site are molded into the precise positions that enable the enzyme to perform its catalytic function. In some cases, such as glycosidases, the substrate molecule also changes shape slightly as it enters the active site. The active site continues to change until the substrate is completely bound, at which point the final shape and charge is determined.

Classification of enzymes:

According to the type of reactions that the enzymes catalyze, enzymes are classified into six categories, which are oxidoreductases, transferases, hydrolases, lyases, isomerases and ligases. Oxidoreductases, transferases and hydrolases are the most abundant forms of enzymes. Individual enzyme classes are further classified systematically based on the chemical name of the substrate and its reaction mechanism.

1. **Oxidoreductases-** catalyze oxidation-reduction reactions where electrons are transferred. These electrons are usually in the form of hydride ions or hydrogen atoms. When a substrate is being oxidized it is the hydrogen donor. The most common name used is a dehydrogenase and sometimes reductase will be used. An oxidase is referred to when the oxygen atom is the acceptor.
2. **Transferases-** catalyze group transfer reactions. The transfer occurs from one molecule that will be the donor to another molecule that will be the acceptor. Most of the time, the donor is a cofactor that is charged with the group about to be transferred. Example: Hexokinase used in glycolysis.
3. **Hydrolases-** catalyze reactions that involve hydrolysis. This cases usually involves the transfer of functional groups to water. When the hydrolase acts on amide,

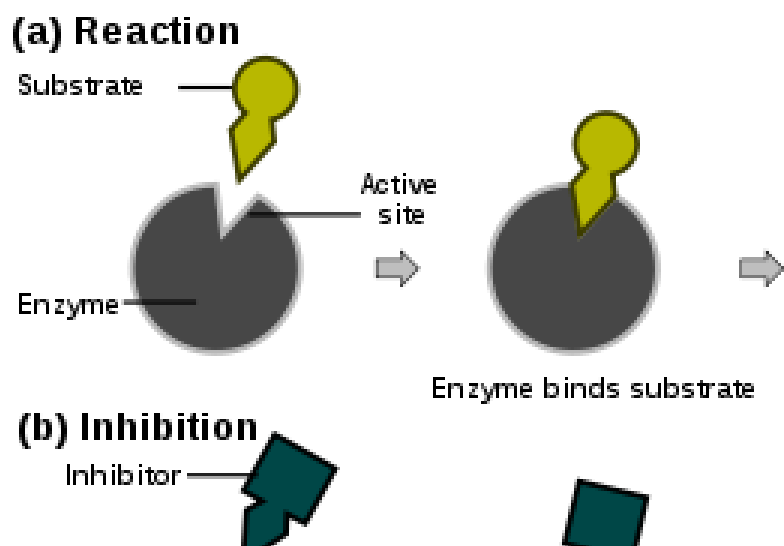
glycosyl, peptide, ester, or other bonds, they not only catalyze the hydrolytic removal of a group from the substrate but also a transfer of the group to an acceptor compound. These enzymes could also be classified under transferases since hydrolysis can be viewed as a transfer of a functional group to water as an acceptor. However, as the acceptor's reaction with water was discovered very early, it's considered the main function of the enzyme which allows it to fall under this classification. For example: Chymotrypsin.

4. **Lyases**- catalyze reactions where functional groups are added to break double bonds in molecules or the reverse where double bonds are formed by the removal of functional groups. For example: Fructose biphosphate aldolase used in converting fructose 1,6-bisphosphate to G3P and DHAP by cutting C-C bond.
5. **Isomerases**- catalyze reactions that transfer functional groups within a molecule so that isomeric forms are produced. These enzymes allow for structural or geometric changes within a compound. Sometime the interconversion is carried out by an intramolecular oxidation-reduction. In this case, one molecule is both the hydrogen acceptor and donor, so there's no oxidized product. The lack of an oxidized product is the reason this enzyme falls under this classification. The subclasses are created under this category by the type of isomerism. For example: phosphoglucose isomerase for converting glucose 6-phosphate to fructose 6-phosphate. Moving chemical group inside same substrate.
6. **Ligases**- are used in catalysis where two substrates are ligated and the formation of carbon-carbon, carbon-sulfide, carbon-nitrogen, and carbon-oxygen bonds due to condensation reactions. These reactions are coupled to the cleavage of ATP.

CLASSIFICATION OF ENZYMES		
Group of Enzyme	Reaction Catalysed	Examples
1. Oxidoreductases	Transfer of hydrogen and oxygen atoms or electrons from one substrate to another.	Dehydrogenases Oxidases
2. Transferases	Transfer of a specific group (a phosphate or methyl etc.) from one substrate to another.	Transaminase Kinases
3. Hydrolases	Hydrolysis of a substrate.	Esterases Digestive enzymes
4. Isomerases	Change of the molecular form of the substrate.	Phospho hexo isomerase, Fumarase
5. Lyases	Nonhydrolytic removal of a group or addition of a group to a substrate.	Decarboxylases Aldolases
6. Ligases (Synthetases)	Joining of two molecules by the formation of new bonds.	Citric acid synthetase

Enzyme Inhibition

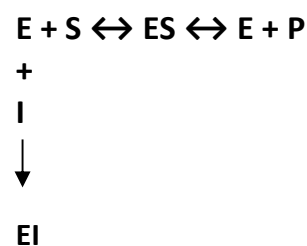
Substances that reduce the activity of an enzyme-catalysed reaction are known as inhibitors. They act by either directly or indirectly influencing the catalytic properties of the active site. Inhibitors can be foreign to the cell or natural components of it. Those in the latter category can represent an important element of the regulation of cell metabolism. Many toxins and also many pharmacologically active agents (both illegal drugs and prescription and over-the-counter medicines) act by inhibiting specific enzyme-catalysed processes. Various types of enzyme inhibitors can decrease enzyme reaction rates.



These inhibitors are of two types, based on their interaction with the enzyme molecule.

1. Irreversible inhibitors

Irreversible inhibitors react with the enzyme and form a covalent adduct with the protein. The inactivation is irreversible. These compounds include eflornithine a drug used to treat the parasitic disease sleeping sickness. Penicillin and Aspirin also act in this manner. With these drugs, the compound is bound in the active site and the enzyme then converts the inhibitor into an activated form that reacts irreversibly with one or more amino acid residues.



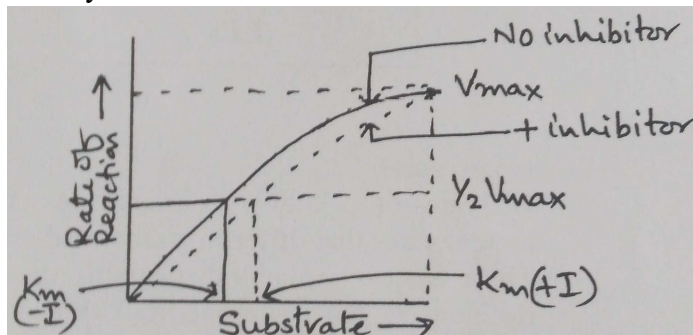
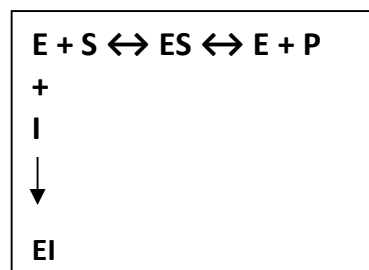
2. Reversible Inhibitors

It involves equilibrium between the enzyme and the inhibitors. They are of three types –

a. *Competitive inhibition*

In competitive inhibition, the inhibitor and substrate compete for the enzyme (i.e., they can not bind at the same time). Often competitive inhibitors strongly resemble the real substrate of the enzyme. For example, methotrexate is a

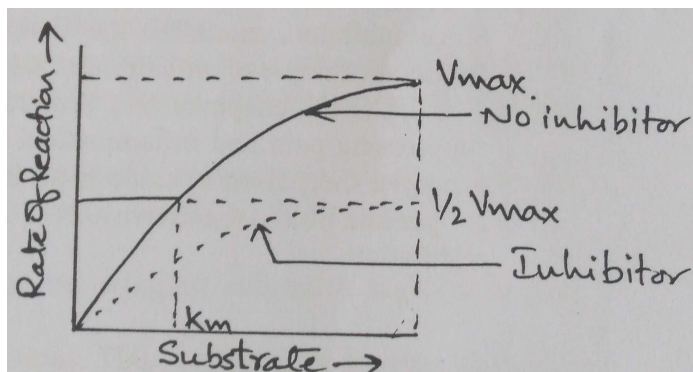
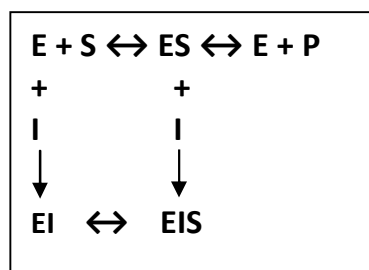
competitive inhibitor of the enzyme dihydrofolate reductase, which catalyzes the reduction of dihydrofolate to tetrahydrofolate.



In this case, ES and EI products are formed but EIS products are never produced. And this condition can be overcome by addition of more substrate. In this case K_m changes but V_{max} will remain same.

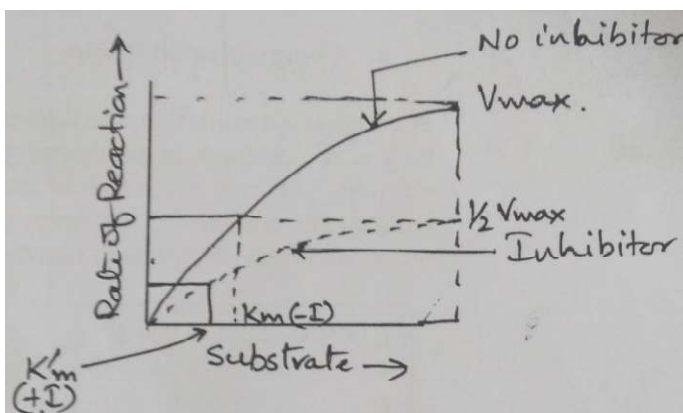
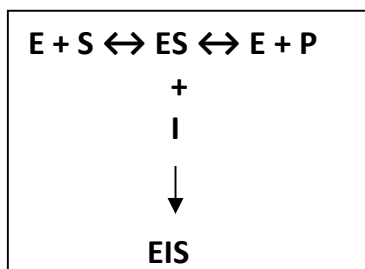
b. Non-competitive inhibition

Non-competitive inhibitors can bind to the enzyme at the binding site at the same time as the substrate, but not to the active site. Both the EI and EIS complexes are enzymatically inactive. Because the inhibitor can not be driven from the enzyme by higher substrate concentration (in contrast to competitive inhibition), the apparent V_{max} changes. But because the substrate can still bind to the enzyme, the K_m stays the same.



c. Un-competitive inhibition

In uncompetitive inhibition, the inhibitor cannot bind to the free enzyme, only to the ES-complex. The EIS-complex thus formed is enzymatically inactive. This type of inhibition is rare, but may occur in multimeric enzymes. In this case both V_{max} and K_m will decrease.



Isoenzymes

Isozymes (also known as isoenzymes or more generally as Multiple forms of enzymes) are enzymes that differ in amino acid sequence but catalyze the same chemical reaction. These enzymes usually display different kinetic parameters (e.g. different K_M values), or different regulatory properties. The existence of isozymes permits the fine-tuning of metabolism to meet the particular needs of a given tissue or developmental stage (for example lactate dehydrogenase (LDH)).

Allosteric enzymes

In a multienzyme sequence reaction some times the end product may act as specific inhibitor for an enzyme at or near the beginning of the sequence. And as a result the rate of entire sequence is determined by the steady state concentration of the end product. This type of inhibition is called the end product inhibition, feed back inhibition or retro inhibition and the enzyme which is inhibited by the end product is called allosteric enzyme.

Uses of inhibitors

Since inhibitors modulate the function of enzymes they are often used as drugs. A common example of an inhibitor that is used as a drug is aspirin, which inhibits the COX-1 and COX-2 enzymes that produce the inflammation messenger prostaglandin, thus suppressing pain and inflammation. However, other enzyme inhibitors are poisons. For example, the poison cyanide is an irreversible enzyme inhibitor that combines with the copper and iron in the active site of the enzyme cytochrome-c oxidase and blocks cellular respiration.